

Hill Country Christian School of Austin

Select Presentation

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Introduction

Mass Shooting Events are more likely to occur in workplaces and schools

- 28% and 22% of recorded events, respectively

Legislative requirements for silent panic button alarms to be installed in public schools exist

- Less effective than a simple 911 call, according to the timeline of some of the most recent school shootings

Technologies that enable faster data exchange and real-time updates significantly enhance the quality of decisions in emergency situations

Problem Statement

Emergency response during rapidly evolving emergencies is hindered by insufficient real-time situational awareness and communication.

Stakeholder Needs

Interpreted Need	Design Application
Communication among Stakeholders	Proposed design should be able to communicate effectively with a diverse group of stakeholders in an emergency (staff, first responders, volunteers).
Information Distribution	Proposed design should include user-facing interfaces that allow for simple and concise communication under high-stress conditions and clearly communicate hazard zones and responder readiness status.
Redundancy and Failover	Proposed design should balance range and latency for optimal real-time response, and maintain reliable coverage across large campuses.
Ease of Integration	Proposed design should support interoperability between civilian teams and law enforcement and integrate easily with existing solutions.
Speed & Accuracy of Information	Proposed design should combine the immediacy of radio with the clarity and data capacity of digital apps while minimizing latency and ensuring accuracy.

System Overview

Mesh-based emergency communication and situational awareness system with three components:

- Bluetooth Low Energy (BLE) room-localization network
- Mobile interface for stakeholders
- Role-based playbooks for users

Designed for environments where

- Traditional security roles may be spread thin
- Volunteers and untrained staff play key roles, and a fluid level of staff training exists

Selection Process

Pugh Chart Evaluation

**Failure Mode & Effects
Analysis (FMEA)**

Pugh Chart #1: User Interface	Handheld LMR radio	Radio + Data	Smartphone App	Body Worn Pack	Voice Controlled Headset
Interoperability	0	1	1	-1	-1
Coverage	0	1	1	1	1
Reliability	0	1	1	-1	-1
Latency	0	1	1	1	1
Duplex Communication	0	1	1	1	-1
Cognitive load	0	0	1	-1	-1
Deployment	0	-1	1	-1	-1
Rich Data	0	0	1	0	0
Hands-Free Operability	0	0	0	1	1
Security	0	-1	1	1	-1
Scalability	0	-1	1	-1	1
TOTAL POSITIVE	0	4	10	5	4
TOTAL NEGATIVE	0	-1	0	-5	-6
TOTAL	0	3	10	0	-2

FMEA Analysis – Mobile Interface

Component/ Function	Potential Failure Mode	Potential Effects of the Failure	Severity	Potential Causes	Probability	Recommended Actions
Detect nearby beacon	Partially degraded function	No location update is made	1	User Settings / Configuration BLE scanning fails/is disabled	3	On-prem devices for volunteers whose devices are incompatible, Startup BLE check, alert if BLE is disabled
Estimate current room	Partially degraded function	Volunteer appears in the wrong place, causing a misdirected response	5	App reads incorrect RSSI / selects the wrong room	2	Use smoothing/filtering, show confidence score, allow manual override
Report location	No function	No location update	5	Data not transmitted (e.g., comms failure)	1	Retry logic, log failed sends, queue updates

Component 1: Mobile Interface

Key components:

- Receive and acknowledge alerts
- Visualize volunteer locations on a floor plan
- Track incident status and assignments
- Differing access levels depending on pre-configured ACL's

Advantages to this solution:

- Multimodal (text, audio, images, maps)
- Most users already know how to use smartphones
- Easily updated and scalable

Pugh Chart #2: Network	Smartphone Application + SOS button	Smartphone Application + BLE Room Locator + SOS	Smartphone Application + BLE Room Locator + Voice-Guided Assistant	Smartphone Application + BLE Room Locator + Task Board
Speed of deployment	0	0	0	0
Technical complexity	0	-1	-2	-1
Infrastructure dependence	0	1	1	1
Cost	0	-1	-1	-1
Training burden / volunteer use	0	1	-1	0
Situational awareness quality	0	2	2	2
Communication clarity under stress	0	1	0	0
Hands-Free Operability	0	0	1	0
Security	0	0	0	1
TOTAL POSITIVE	0	5	4	4
TOTAL NEGATIVE	0	-2	-4	-2
TOTAL	0	3	0	2

FMEA Analysis – BLE beacons

Component/ Function	Potential Failure Mode	Potential Effects of the Failure	Severity	Potential Causes	Probability	Recommended Actions
Transmit beacon signal	No function	Beacon stops transmitting, Current Room cannot be detected	3	Overuse, lack of maintenance	5	Use long-life batteries, use a battery status indicator, tie into PoE lines
Broadcast room ID	Intermittent function	The system can't map the signal to a room, causing location errors	3	Corrupted Firmware, Device ID corrupted / reset	1	Lock firmware, verify ID in app, label devices clearly with physical room ID
Maintain signal strength	Intermittent function	Signal is too weak/inconsistent, wrong room estimated	1	The beacon is placed behind an obstruction or in the wrong spot	2	Establish standard deployment positions, test signal strength during install

Component 2: BLE Room-Localization Network

BLE beacons installed at the room level throughout a campus

- Beacons broadcast room IDs and health/status information
- User's mobile devices or dedicated handhelds scan for nearby beacons during an emergency event, and estimate current room based on signal strength and ID mapping

Benefits:

- Room-level accuracy without GPS, can be supplemented through WiFi BSSID scanning
- Works in complex, signal hostile, or multi-building campuses
- Supports real-time location tracking of volunteers and staff to determine where an incident has occurred, and who is the closest to respond.

Component 3: Role-based playbooks

Embedded role-specific playbooks and checklists inside the app

Supports:

- Area captains, security, classroom teachers, and volunteers
- Clear, step-by-step actions during escalating events

Designed for offline operation:

- Critical procedures cached on the device
- Can function during network or power outages

Reduces cognitive load by replacing free-form radio chatter with structured, scenario-based guidance

How are Stakeholder Needs Met?

Interpreted Need	How does the design meet this need?
Communication among Stakeholders	- The proposed design meets this need through the implementation of a multimodal mobile application that supports different viewpoints and access levels depending on organization position.
Information Distribution	- The proposed design meets this need through the implementation of a multimodal mobile application that supports different viewpoints and access levels depending on organization position.
Redundancy and Failover	- The proposed design balances range and latency with room-to-room localization with the integration of a BLE mesh localization system.
Ease of Integration	- Because the system is built on relatively low-cost hardware and software, it can be scaled to different campuses or organizations with varying budgets
Speed & Accuracy of Information	- The proposed design enables the rapid sharing of information in an infrastructure-hostile environment

Q&A
